Shikhar Shiromani

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EXPERIENCE

Apple *June 2026 - Present*
Senior ML Engineer
- Developing intelligent systems that optimize power and performance across Apple's operating systems

NVIDIA *Feb 2025 - June 2026*
Senior Software Engineer - Tech Lead
- Built the first server lifecycle management system at NVIDIA that streamlines resource requests, allocations, and deployment tracking for faster data center rack bring-up
- Explored ML techniques to enhance node uptime by predicting failures and optimizing system reliability

NVIDIA *May 2024 - Aug 2024*
Machine Learning Research Intern
- Architected an ML-driven predictive maintenance system for over 1000 Blackwell/Hopper nodes, reducing downtime and delivering over **$100k/year** in operational savings
- Implemented the model for multidimensional multimodal anomaly detection with negative-sampling and integrated gradients for variable attribution during model interpretation.
- Developed microservices for telemetry collection (6200+ metrics in 2 mins), dashboarding (Grafana) and failure prediction
- Deployed an MLOps pipeline using Airflow and implemented a FastAPI based REST service, handling over 10,000 inferences/day
- Presented a poster at NTech Conference 2024. Filed for patent (first author).

AI Institute for Advances in Optimization (AI4OPT) *Aug 2023 - Dec 2024*
Graduate Research Assistant (Georgia Tech), advised by Prof. Pascal Van Hentenryck
- Founding backend engineer responsible for developing a highly available backend system for an on-demand multi-modal transportation solution, focusing on large-scale ride-sharing. Available on iOS and Android.
- Successfully piloted in Savannah, GA over 15 months to schedule more than 100 rides daily
- The project was awarded one of the SMART 20 Awards 2025

NVIDIA *Jul 2019 - Jul 2023*
Software Engineer II
- Architected and wrote a scalable self-serve CI/CD platform, cutting engineering workload turnaround time by 93%
- Presented a paper at NTech India Conference, discussing several benefits of the system
- Delivered the NvCCP (NVIDIA's Camera Control Protocol) module for successful automotive safety assessments by driving the software architecture, features and functional verification of the module

RESEARCH

- **Pivotal Research Fellow - CoT Faithfulness in LLMs** *Feb 2026 – Present*
  - Selected as a Research Fellow from 2,500+ applicants
  - Working with Noah Y. Siegel (Google DeepMind, UK) on CoT faithfulness in LLMs
- **MARS V Research Fellow - Steganographic Reasoning in LLMs** *July 2026 – Present*
  - Working with Julian Schulz on showing LLMs hide CoT in text via a distributed mechanism that transfers across encoding schemes by reusing the same weights
- **Benchmarking Pluralistic Alignment Through Persona-Conditioned Behavioral Evaluation (ICML 2026 Pluralistic Alignment Workshop)** *June 2026*
  - Built the Pluralistic Alignment Benchmark (PAB), covering 320 scenarios, 8 behavioral-risk categories, and 63 synthetic personas
  - Evaluated 9 leading LLMs with a multi-judge pipeline, revealing substantial persona-conditioned drift in manipulation, concealment, persuasion, and sycophancy
- **Visuals Lie, Consistency Speaks in VLMs (ICLR 2026 MM Intelligence Poster)** *Dec 2025 - Mar 2026*
  - Built VLM Reliability Probe, a cross-family reliability probe testing whether spatial attention structure predicts VLM correctness
  - Showed attention coherence has near-zero correlation with accuracy, while self-consistency is a much stronger reliability signal

- **SAGE: Agentic Framework for Interpretable Pathology Biomarker Discovery** *Jul 2025 - Present*
  - Developed an agentic framework for discovering pathology biomarkers grounded in biological literature and multimodal evidence
  - Linked image-derived features with molecular markers and clinical outcomes to improve transparency and clinical translatability
- **ProMoral-Bench: Evaluating Prompting Strategies in LLMs (NeurIPS 2025 PersonaNLP)** *May 2025 – Nov 2025*
  - Built a benchmark comparing 11 prompting strategies across 4 LLM families, including the new ETHICS-Contrast dataset
  - Introduced the Unified Moral Safety Score and showed exemplars outperform multi-hop reasoning in accuracy, safety, and efficiency
- **ChameleonBench: Quantifying Alignment Faking in LLMs (ACML 2025 (Oral))** *May 2025 – Jun 2025*
  - Built an 800-prompt benchmark across 8 misbehavior categories with an external judge pipeline for alignment-faking evaluation
  - Evaluated 6 leading LLMs, showing up to 20% higher harmful response rates in free-form versus test-like settings
- **The Hypocrisy Gap: Detecting Divergence Between Internal Belief and CoT via Sparse Autoencoders** *Jan 2026*
  - Introduced a mechanistic metric to quantify divergence between internal model belief and final generated explanation
  - Outperformed a decision-aligned log-probability baseline for detecting sycophancy across model families
- **Linear Predictability of Attention Heads in LLMs** *Aug 2024 – May 2025*
  - Revealed emergent linear relationships among attention heads, enabling cross-head approximation
  - Compressed KV cache through learned projections, achieving up to 2× reduction with minimal performance loss
- **Context-Modulated PID Attention Steering for Hallucination Mitigation (AAAI 2026 LaMAS Oral)** *Nov 2025*
  - Designed an interpretable feedback-driven decoding framework that steers attention using the Context Reliance Score
  - Reduced contextual hallucinations by **2.8–5.8%** across HotpotQA, XSum, HaluEval, and RAGTruth without retraining

EDUCATION

Georgia Institute of Technology, Atlanta, USA *Aug 2023 - Dec 2024*
Master of Science in Computational Science and Engineering, GPA: 4.0/4.0

Birla Institute of Technology and Science, Pilani, India *Aug 2015 - Jun 2019*
Bachelor of Engineering in Electrical and Electronics Engineering, GPA: 8.20/10.0

SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, SQL
Tools/Frameworks: PyTorch, Django/Flask, NodeJS, MySQL, InfluxDB, K8s, Docker, ReactJS, Jenkins
Coursework: Systems for ML, Deep RL, High Performance Computing, Database Systems Implementation, Algorithms